Rubber Hose Revival: The DreamWorks Curvy-Limb Rig

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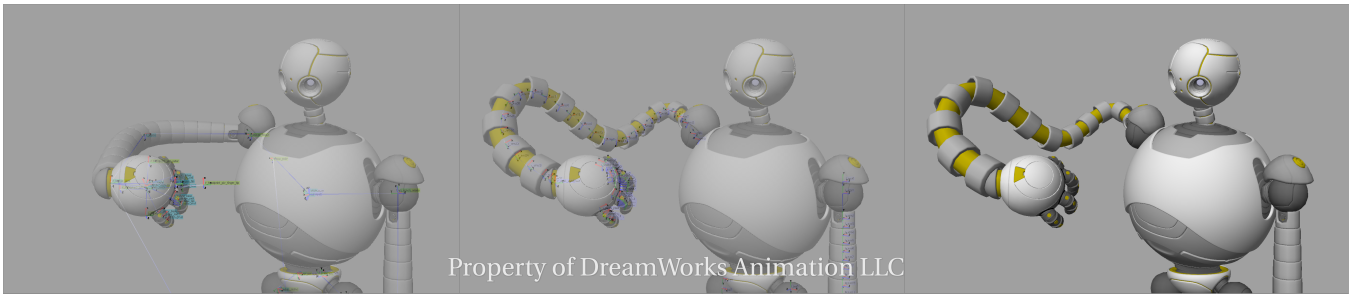


Figure 1: The DreamWorks Curvy-Limb rig resamples limb elements into high resolution joint splines and adds controls to more easily pose curved shapes. (Left: Arm skeleton posed with traditional arm controls. Center: Resampled joint spline posed with the Curvy-Limb controls. Right: Final deformation result.)

Abstract

The DreamWorks Curvy-Limb rig allows animators to pose complex curves without sacrificing intuitive usability. The system adds additional flexibility to character limbs (arms and legs) and digits (fingers and toes) while retaining the standard arm, leg, and digit controls to which animators are already familiar. Animators can activate the extra controls as needed on the fly without having to blend into any special "states." The robustness of the Curvy-Limb has proven to be instrumental to the character performances of multiple DreamWorks Animation productions, including *Ruby Gillman, Teenage Kraken*, *Trolls Band Together*, and *The Wild Robot*.

CCS Concepts

• **Computing methodologies** → **Animation**.

Keywords

curvy limb, rubber hose, bendy, ruby gillman, wild robot, trolls band together, animation, rig

*Worked at DreamWorks Animation during the production of *Ruby Gillman, Teenage Kraken*

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1 Introduction

During production of *Ruby Gillman, Teenage Kraken*, a new control system was required to support the flexibility of the "boneless" bodies of the kraken characters. While it is well known how complicated it can be to rig and efficiently animate flexible curved structures such as tentacles, snakes, and ropes, this production had the added complexity of those same structures needing to double as traditional limbs [Boucher and Dirksen 2015; Hessler and Talbot 2016; Inagaki et al. 2022; Sheerin et al. 2022; Stevenson and Bernard 2024; Talbot et al. 2016]. Before reinventing the wheel, useful techniques and features of complex character rigs like snakes and octopusses from previous DreamWorks films were explored [Boucher and Dirksen 2015; Inagaki et al. 2022]. These systems included layered deformation systems, existing motion systems like our modular "tail" and "tentacle" rigs, and even a control to blend the arm and leg skeleton to that of a tentacle. Simply layering lots of separate controls became more difficult for animators to manage without much internal rig logic on how they relate to one another. With the hybrid arm-tentacle blend approach, animators could not pose the advanced tentacle controls when in "arm" mode, severely limiting performance possibilities. Our traditional approach to "bendy limbs" was also limited. The system is implemented by layering a secondary curve deformer on the character after being deformed by the base skeleton. Since this is only an offset on the underlying deformation, there are limitations in the ability to create smooth shapes across limb joints if the underlying

shape is not already smooth. Since an entire family of primary characters required these controls for a feature production, it was clear something much more intuitive was needed, leading to the creation of the Curvy-Limb.

2 Methodology



Figure 2: The Curvy-Limb rig provides a familiar and intuitive interface for animators.

2.1 Design Philosophy. The Curvy-Limb is a layered motion system that takes advantage of our procedural rigging graph to resample the skeleton of an arm or leg rig into a high resolution spline (Figure 1) [Hutchinson et al. 2019]. Animators can use additive "handle" controls to shape the tangents of the curve as it runs through the input joints (which work like anchor points on a spline). 3D Animators are already accustomed to manipulating splines and tangents when adjusting animation curves, and this rig interface was designed to capitalize on that familiarity (Figure 2). The new "curvy" controls follow the traditional arm and leg motion systems. This way, animators could use familiar arm and leg rigs, and choose to add extra curvature where needed. Since the "curvy" skeleton is dynamically resampled from the arms and legs, deformation from the "curvy" controls is offset in a space that is intuitive to the user, respecting rotation, twist, and scale of the underlying limbs. Animators could repose an arm without having to completely repose the complex curvature on top. This system also simplified the rigging process by creating a unified system for both the arms and legs. Updates to the system during production could be made in one rebuildable motion package. The procedural nature of the system also meant rigging artists did not have to place any new joints. They simply had to place the joints of a standard skeleton. Therefore once the system was set up, it essentially didn't add any rig setup time when starting new characters.

2.2 Core Features. Offset controls on the elbow or knee drive the system. A "curviness" control determines the default middle tangent curvature when bending the limb. This controls if the default elbow/knee bend has a broad "U" or a sharper "V" shape. Additionally, there is a separate "sharpness" control which moves the mid-joint's tangent handles closer or further away from the mid joint. "Sharpness" and "aim" controls exist for the handles at the beginning and end of the spline (shoulder/hip and hand/foot) as well. In addition to all these controls determining the inherent behavior of the limbs, animators could also move every handle directly in 3D space. Features inspired by our "tentacle" rig were then layered onto this functionality [Boucher and Dirksen 2015]. Secondary offset controls that ride along the curve have "sharpness" and "location" controls to determine where and how much of the curve they affect. This gives an effect similar to what is sometimes called "Variable FK" [Stevenson and Bernard 2024]. The Curvy-Limb also includes decoupled "scale" controls so animators wouldn't be limited by the handle positions for where they can scale. Controls were added to twist the entire limb or sections of the limb along the curve. There are other controls to help manage volume preservation when stretching a limb. And finally, slideable "parent" joints were added that allowed animators to constrain objects to the limbs and slide their location along the curve.

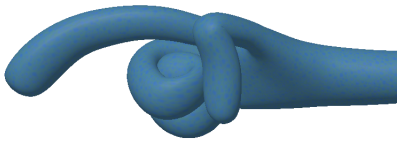


Figure 3: The Curvy-Limb rig has an "as digit" mode to adapt it for use for fingers or toes.

2.3 Advanced Features. The character designs for *Ruby Gillman, Teenage Kraken* required hand poses with clean spirals (Figure 3). This led to an additional "as digit" mode for the Curvy-Limb. The Curvy-Limb was modular, but was designed specifically for motion systems with a clear middle "hinge." For the new "as digit" mode, the structure of the elbow/knee controls had to be adapted to the multiple knuckles of a finger or toe. To simplify the number of controls and icons on a full hand, in this mode, the translation controls of the individual knuckle handles are removed, and rely on the existing "curviness" and "sharpness" controls instead. Our modular tentacle and tail rigs have a built-in "wave" functionality to allow animators to control amplitude, frequency, and phase of a wave that moves down the appendage. This system was broken out as a separate motion package, so it could be easily added as a post-process to the Curvy-Limb. This new "Wave" package could be used elsewhere, such as layering it on a four joint FK chain to quickly rig a simple fish. A "slide" feature was also added to the Curvy-Limb, allowing an animator to "grow" or "retract" a limb along its curve. This was achieved by intercepting the end parameter value of the up-resed spline curve with an animation control that changes the end position of the spline dynamically, giving the effect of the limb following a path.

3 *The Wild Robot*

While *Ruby Gillman, Teenage Kraken* and *Trolls Band Together* used the Curvy-Limb for characters with designs reminiscient of classic "rubber hose" style animation of the 1920s and 1930s, *The Wild Robot*

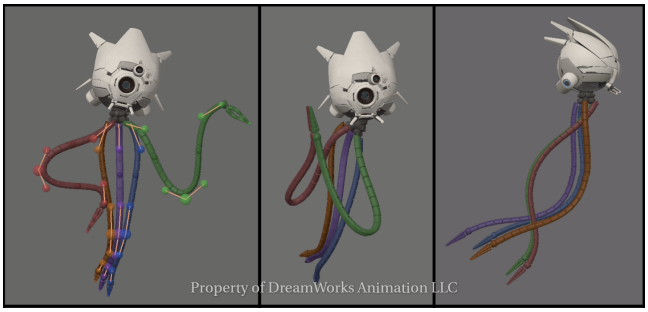


Figure 4: For Vontra, a "Color Debug" mode was added to the limbs to help animators keep track of tangled tentacles throughout a shot.

tested the system further on hard-surface robots. The character *Roz*, has a series of metal rings that overlap a streching central tube for her limbs. These rings needed to keep their rigid structure while the underlying tube stretched. This was achieved by using the Curvy-Limb package options to match the number of up-resed spline joints to the number of geometric rings, and rigidly binding the geometry to those joints. The arc length of each scaled limb segment was then calculated, and the limb segments were adjusted to scale evenly, allowing the rings to be evenly distributed across the stretched limbs. The villain robot, *Vontra*, resembles a squid with five free-floating tentacles. Instead of using the robust but arduous "tentacle" rig from past shows, we used the Curvy-Limb system to get the shapes we needed. This kept animator interaction intuitive by treating each tentacle as an arm, rather than managing multiple complicated IK splines [Boucher and Dirksen 2015]. Additional controls were then added to *Vontra* to help manage the multiple tentacles. Since *Vontra's* arms weren't constrained to bend in only one direction, controls were added to reverse the bend direction of the underlying limb IK, so animators could decide if the default bend of the tentacle "elbow" was inward or outward. When animating the overlapping black tentacles, it became difficult to track them over the length of a shot. A "Color Debug" switch was added to color code the legs and control icons, making it easier to see which limb was which (Figure 4). A "Primary" IK jack was also added to control all of the tentacles together. This included blend controls to turn off the functionality per limb.

4 *Challenges*

While the rigs at DreamWorks are highly parallelized, there is a step in the rig graph that combines all the joints together to process the skeleton for downstream departments such as crowds [Watt et al. 2012; Zimmermann et al. 2014]. The more joints there are, the slower it is to evaluate. On a typical character rig the effect is negligible. However, when there are multiple characters in a shot where each limb has 50 joints and each finger and toe has 25 joints, it starts to have an impact. The limb "slide" system created more upstream dependencies which exacerbated the issue. In a "curvy" character, all the Curvy-Limb systems installed were evaluating in parallel. This caused an issue when using the "slide" or "wave" controls, where the fingers were left behind, tearing the hand geometry. This is because the upstream inputs of the "curvy" fingers were the traditional digit motion systems witout any inheritance from the "curvy" arm. If the inputs to the basic digits were changed to use the "curvy" arm, heavier rig bottlenecks would occur, and key inputs for functionality like the "hand-rocker" would be missing. Instead, offsets from the pre and post "slide" and "wave" poses of the arm were dynamically calculated and applied to the digits as a post-process, so fingers could be affected by the slide while still calculating all the Curvy-Limb systems mostly in parallel. Using the slide system also created an awkward state for animators where the controls of the arm could no longer be used to shape the retracted arm. Posing the arm controls meant reshaping the curve path that the retracted limb was following. A number of additional "slideable" FK controls were added "post-slide" in the rig graph to give some extra control for when this feature was used.

5 Conclusion



Figure 5: The Curvy-Limb rig being used on various DreamWorks Animation productions. (Left to Right: *Ruby Gillman, Teenage Kraken* (2023), *The Wild Robot* (2024), *Trolls Band Together* (2023), *The Wild Robot* (2024))

The versatility of the DreamWorks Curvy-Limb rig has been widely embraced by the animation department. On *Ruby Gillman, Teenage Kraken* it was used for the entire Gillman family, their pet, *Nessie*, the multi-legged giant kraken characters (*Grandmama* alone had 2 arms, 6 legs, 8 fingers, and 24 toes, all with Curvy systems), and was even put through the crowds pipeline for the background kraken characters. The system worked as expected for characters with different proportions and limb thicknesses. The system was next used on *Trolls Band Together* for the villains *Velvet* and *Veneer*, as well as the crowds of "*Mount Rageons*" and "*Vacay Islander*" characters. Most recently *The Wild Robot* showed how the same system was robust enough to be adapted for hard-surface robot characters (Figure 6). The use of the Curvy-Limb for *Vontra* poetically proved there could be a simpler way to animate complicated tentacles. Thanks to this system, animators spent less time managing keys, and did not need to completely re-pose dozens of offsets in constantly changing spaces when addressing director notes. There is interest in adapting the control interface to be even more modular, to add curvature to other features, such as spines, necks, and various props.

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